

E-Commerce Customer Retention & Sales Analysis

Project Documentation

Tools: SQL, Python, Power BI, Power Query, DAX

Dataset: Olist Brazilian E-Commerce Dataset

Project Type: Data Analytics / Business Intelligence / Customer Retention

Project Objective

Analyze e-commerce sales performance, customer behavior, product performance, seller performance, and customer retention opportunities using SQL, RFM analysis, and an interactive Power BI dashboard.

Prepared as a Data Analyst Portfolio Project

1. Project Title

E-Commerce Customer Retention & Sales Analysis

Analyzing sales performance, customer behavior, product categories, seller performance, and customer retention using SQL and Power BI.

2. Project Overview

This project analyzes an e-commerce dataset to understand overall business performance and customer purchasing behavior. The analysis combines SQL-based data analysis, RFM customer segmentation, and an interactive Power BI dashboard.

The analysis covers:

- Sales and revenue trends
- Customer purchasing patterns
- One-time vs repeat customers
- Customer retention opportunities
- High-value customer segments
- Top-performing product categories
- Seller and geographic performance
- Payment-method behavior

The final Power BI report contains four analytical pages:

- Executive Sales Overview
- Customer & Retention Analysis
- Product & Sales Performance Analysis
- Executive Summary

3. Business Problem

The e-commerce business generates a large volume of orders, but simply increasing sales is not enough for long-term growth. The business needs to understand revenue drivers, customer behavior, repeat purchasing, product performance, and retention opportunities.

Main Business Problem: How can the e-commerce business use customer and sales data to improve revenue performance, increase repeat purchases, and strengthen customer retention?

4. Project Objectives

Sales Analysis

- Calculate total revenue and orders.
- Analyze monthly revenue trends.
- Calculate Average Order Value.
- Identify top-performing product categories.

Customer Analysis

- Calculate total customers.
- Identify one-time and repeat customers.

- Calculate repeat customer rate.
- Analyze customers by geographic location.

Customer Retention

- Perform RFM analysis.
- Segment customers based on purchasing behavior.
- Identify loyal, champion, potential, and at-risk customers.
- Determine revenue contribution by customer segment.

Product & Seller Analysis

- Identify top product categories.
- Analyze item volume by category.
- Identify top seller cities.
- Compare product revenue and sales volume.

Business Intelligence

- Build an interactive Power BI dashboard.
- Add slicers for filtering analysis.
- Present important KPIs and business insights in an executive-friendly format.

5. Dataset Description

The project uses the Olist Brazilian E-Commerce Dataset, containing information about Brazilian e-commerce transactions and related customer, product, seller, payment, review, and geographic data.

Dataset Table	Description
Customers	Customer information and location
Orders	Order details and timestamps
Order Items	Products purchased in each order
Products	Product information
Payments	Payment methods and payment values
Reviews	Customer review information
Sellers	Seller information and location
Geolocation	Brazilian ZIP-code geographic information
Category Translation	Portuguese-to-English product category mapping

Approximate dataset size: 99K customers/orders, 113K order items, 33K products, 104K payment records, 99K reviews, 3K sellers, and 1M+ geolocation records.

6. Data Cleaning & Preparation

Database Setup

The ecommerce_analysis database was selected and prepared for analysis.

Staging Tables

Separate staging tables were used for customers, orders, order items, products, payments, reviews, sellers, geolocation, and category translation.

Data Loading

Source datasets were imported into MySQL staging tables.

Data Validation

Row counts, sample records, joins, and analytical outputs were checked after loading.

Data Transformation

Dates were prepared for time analysis; customer, order, item, product, and category tables were joined; revenue and customer metrics were calculated.

Power BI Preparation

Cleaned and structured tables were connected to Power BI and relationships were established for reporting.

7. SQL Analysis

SQL was used as the primary analytical layer before visualization. The analysis included overall sales, monthly revenue, product categories, geographic performance, customer value, repeat customer behavior, and RFM segmentation.

Overall Sales KPIs

KPI	Result
Total Revenue	■13.59M
Total Orders	99,441
Total Customers	~96K
Total Items Sold	112,650
Average Order Value	■136.68
Total Freight	~■2M

Monthly Revenue Analysis

Monthly revenue was calculated by grouping orders according to purchase month. November 2017 recorded the highest monthly revenue at approximately ■1.01M.

Product Category Analysis

Revenue was calculated by joining order items with products and category translation. Health & Beauty was the strongest revenue-generating category at approximately ■1.26M.

Geographic Analysis

Customer and sales performance were analyzed by Brazilian state and seller city. São Paulo was the strongest geographic market in the dashboard.

Repeat Customer Analysis

Customers were classified using their number of orders associated with customer_unique_id. Approximately 93K customers were one-time buyers and approximately 3K were repeat customers.

8. RFM Methodology

RFM stands for Recency, Frequency, and Monetary value. It is used to understand customer engagement and identify valuable customer segments.

Metric	Meaning
Recency	How recently the customer purchased
Frequency	How often the customer purchased
Monetary	How much the customer spent

RFM Score = R Score + F Score + M Score. Customers were scored from 1–5 for each dimension and segmented according to the combined score.

RFM Score	Segment	Customers	Revenue
12–15	Champions	12,031	■3.61M
9–11	Loyal Customers	43,604	■7.24M
6–8	Potential Customers	34,346	■2.56M
≤5	At Risk / Lost	5,439	■0.18M
Total		95,420	■13.59M

9. Power BI Dashboard

Page 1 — Executive Sales Overview

KPIs: Total Revenue, Total Orders, Total Customers, Average Order Value, Total Items Sold, Total Freight, and Repeat Customer Rate. Visuals include customer retention, monthly revenue, top sellers, product categories, payment revenue, and key insights.

Page 2 — Customer & Retention Analysis

KPIs: Total Customers, One-Time Customers, Repeat Customers, Repeat Customer Rate, and RFM Customers. Visuals focus on RFM segments, customer distribution by state, revenue contribution by segment, and retention insights.

Page 3 — Product & Sales Performance Analysis

KPIs: Total Revenue, Total Products, Total Items Sold, Average Order Value, and Total Orders. Visuals analyze monthly revenue, product categories, seller cities, category performance, and key insights.

Page 4 — Executive Summary

A concise management-level view of overall revenue, orders, customers, AOV, items sold, repeat customer rate, monthly revenue, top categories, top seller cities, and retention.

10. Key Insights

- **Customer retention is the biggest opportunity:** only 3.12% of customers are repeat customers, while approximately 93K are one-time buyers.
- **Loyal customers generate significant revenue:** approximately 43.6K Loyal Customers contribute around ■7.24M.
- **Potential Customers represent a major opportunity:** approximately 34.3K customers could be targeted for second-purchase conversion.
- **Champions should be protected:** approximately 12K Champions contribute around ■3.61M.
- **Health & Beauty is the strongest revenue category:** approximately ■1.26M.
- **São Paulo is the strongest market:** it leads the geographic performance shown in the dashboard.
- **November 2017 had the strongest monthly revenue:** approximately ■1.01M.
- **Credit cards dominate payment value:** approximately ■12.54M in payment value is shown for credit cards.

11. Business Recommendations

Improve Customer Retention

Use personalized email campaigns, post-purchase follow-ups, second-purchase coupons, loyalty rewards, and product recommendations.

Target Potential Customers

Prioritize the approximately 34K Potential Customers with personalized offers designed to generate a second purchase.

Build a Loyalty Program

Provide Champions and Loyal Customers with exclusive discounts, early access, free shipping, loyalty points, and personalized offers.

Protect High-Value Customers

Monitor Champions for declining purchase activity and use proactive retention campaigns.

Increase Cross-Selling

Use high-performing categories such as Health & Beauty to recommend complementary products.

Use Seasonal Campaigns

Investigate the November revenue peak and prepare stronger campaigns around high-performing periods.

Focus on Strong Geographic Markets

Optimize marketing, logistics, seller partnerships, and inventory availability in high-performing locations such as São Paulo.

12. Tools & Technologies

Technology	Purpose
MySQL / SQL	Data loading, validation, joins, aggregation, customer analysis, revenue analysis, RFM segmentation
Python	Data preparation and exploratory analysis workflow
Power Query	Data transformation and preparation
Power BI	Interactive dashboards, KPIs, charts, slicers, reporting
DAX	Measures, KPIs, revenue calculations, customer metrics, repeat customer rate, AOV

13. Project Outcome

The project transformed raw e-commerce transaction data into an interactive business intelligence solution. The analysis identified key revenue drivers, customer segments, geographic trends, product performance, and customer-retention opportunities.

Final business metrics:

■13.59M revenue | 99K orders | ■136.68 AOV | 3.12% repeat customer rate

The RFM analysis identified 43.6K Loyal Customers, 34.3K Potential Customers, 12K Champions, and 5.4K At Risk / Lost customers. The analysis indicates that customer retention is the biggest growth opportunity, while Loyal Customers and Champions represent important sources of revenue.

14. Final Project Statement

This project demonstrates an end-to-end data analytics workflow, from raw e-commerce data preparation and SQL-based analysis to RFM customer segmentation and interactive Power BI reporting. The analysis identified key

revenue drivers, customer segments, geographic trends, and significant customer-retention opportunities. The resulting dashboard enables business stakeholders to monitor performance and make data-driven decisions to improve customer retention and long-term revenue growth.